

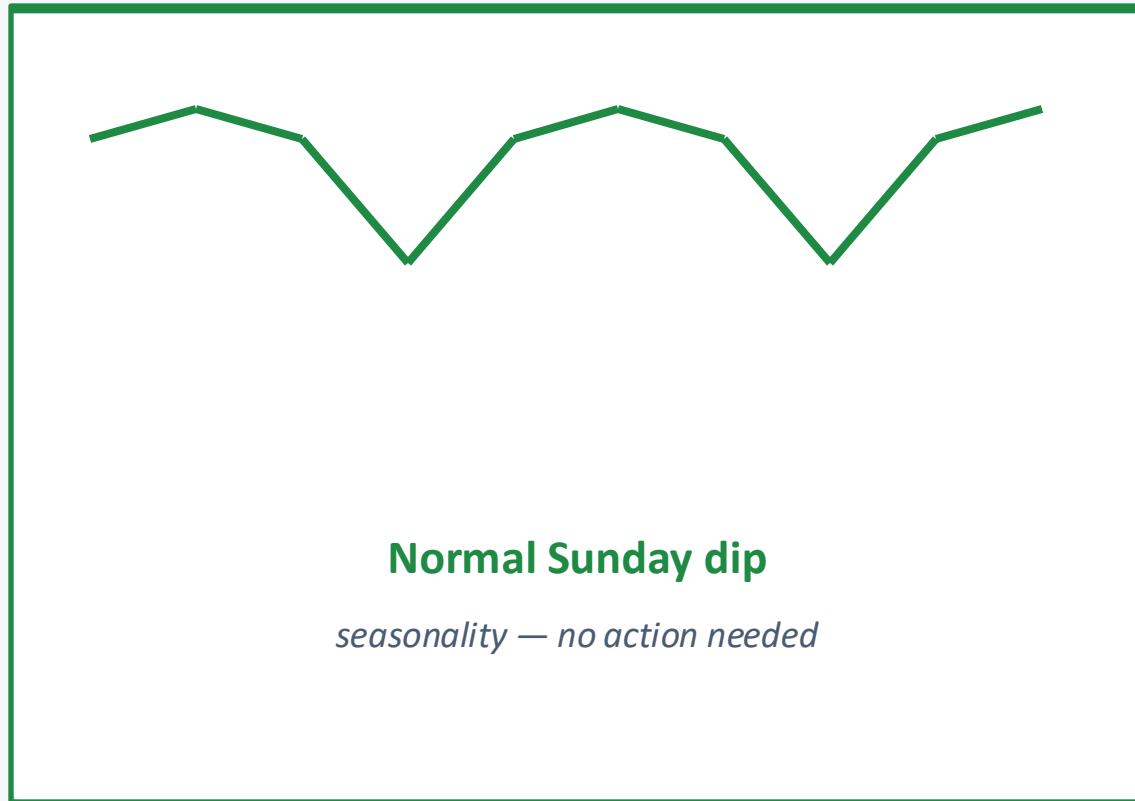


Anomaly Detection

When a metric moves, is it Sunday — or an incident?

Two dips. Only one is an incident.

The detection layer separates them automatically, every hour.

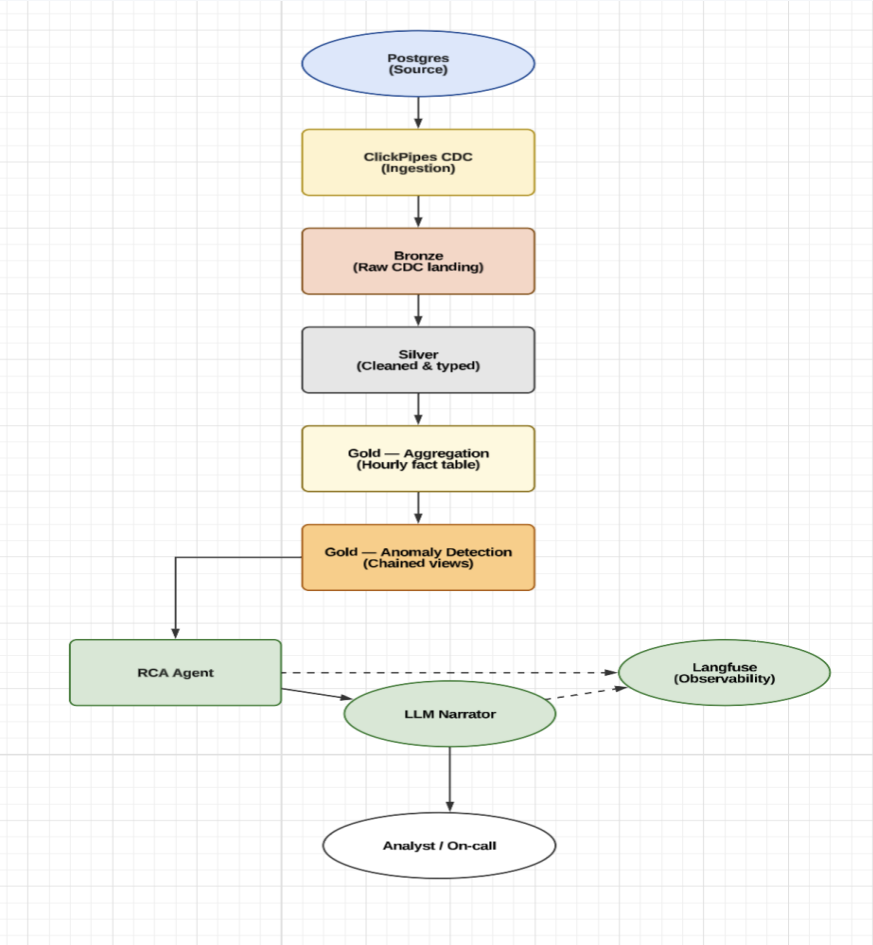


vs.

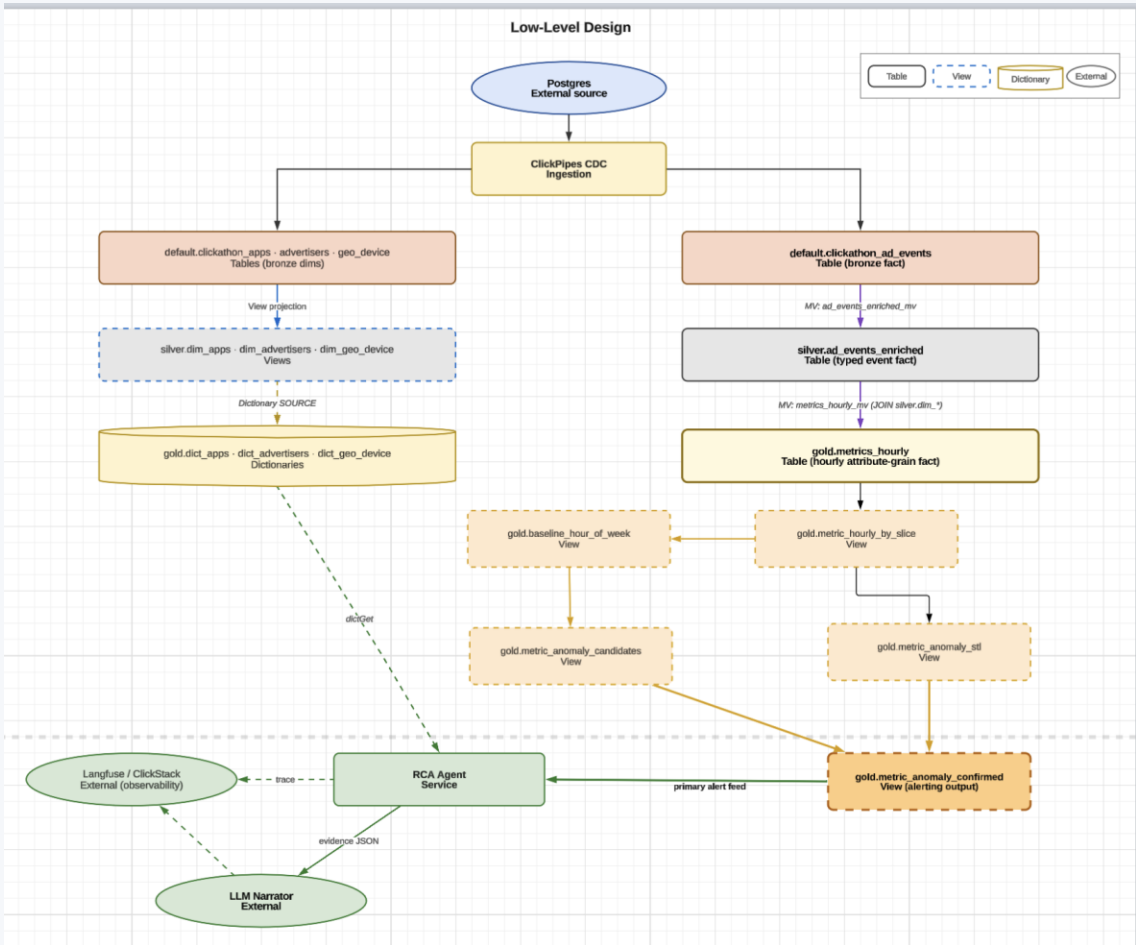


How it's built — HLD and LLD.

HLD

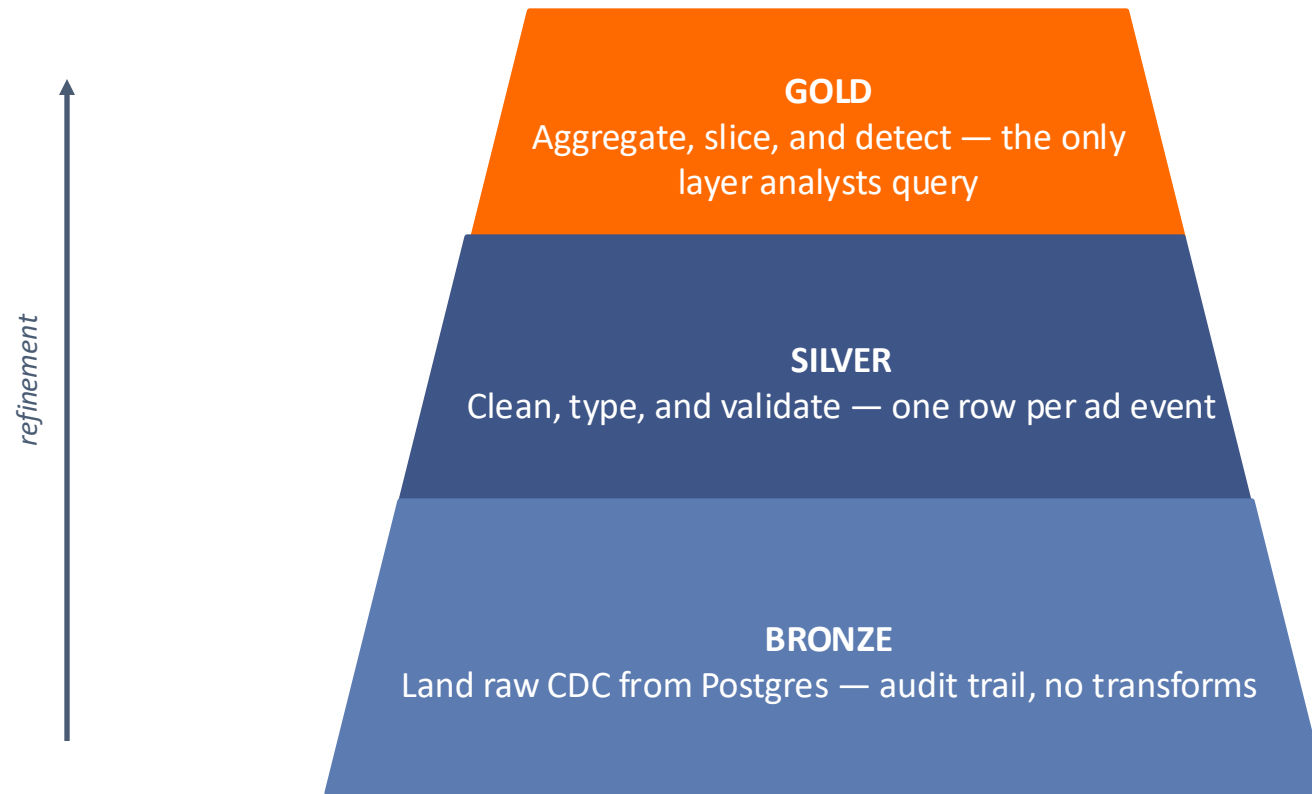


LLD



Medallion architecture.

Three layers. One job each. No overlap.



Postgres → ClickPipes → bronze.* → silver.* → gold.*

Zero-touch pipeline.

Two mechanisms. No batch jobs.

1

MVs move data layer to layer

Every Postgres insert auto-flows Bronze → Silver → Gold.
No scheduler. No click.

```
insert → ad_events_enriched_mv → metrics_hourly_mv
```

2

Views detect anomalies on read

Gold views recompute against the latest hour. Anomalies
surface with zero human trigger.

```
gold.metrics_hourly → metric_anomaly_confirmed
```

Data in → anomalies out. Fully automatic.

How the layer thinks.

Compare today to the same slot last week. If it's off, investigate — five deterministic steps.



Same DoW · same hour · last 14 days.

- 1

DETECT
“Is today outside the normal band?”
- 2

DECOMPOSE
“Which factor moved? Requests · Fill · eCPM · CTR”
- 3

LOCALISE
“Which segment? Region × Format · Region × Vertical”
- 4

RULE OUT
“Seasonality? Ingest event? Cold-start? Low-volume noise?”
- 5

NARRATE
“LLM writes the story from the evidence.”

How we decide it's a real anomaly.

Two rules trigger a candidate. Three independent methods confirm it.

Rule 1 · History disagrees
Rule Z · $|z| \geq 4 \cdot |\Delta\%| \geq 5\%$

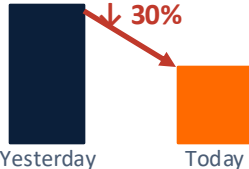


Today is far outside the last 3 same-hour readings.

e.g. fill rate normally $78 \pm 0.3\%$. Today: 72%

- ▶ baseline: 14d trailing · same DoW × HoD · leave-one-out 24h

Rule 2 · The move is too big to ignore
Rule R · $|\Delta\%| \geq 20\% \cdot \text{volume-gated}$

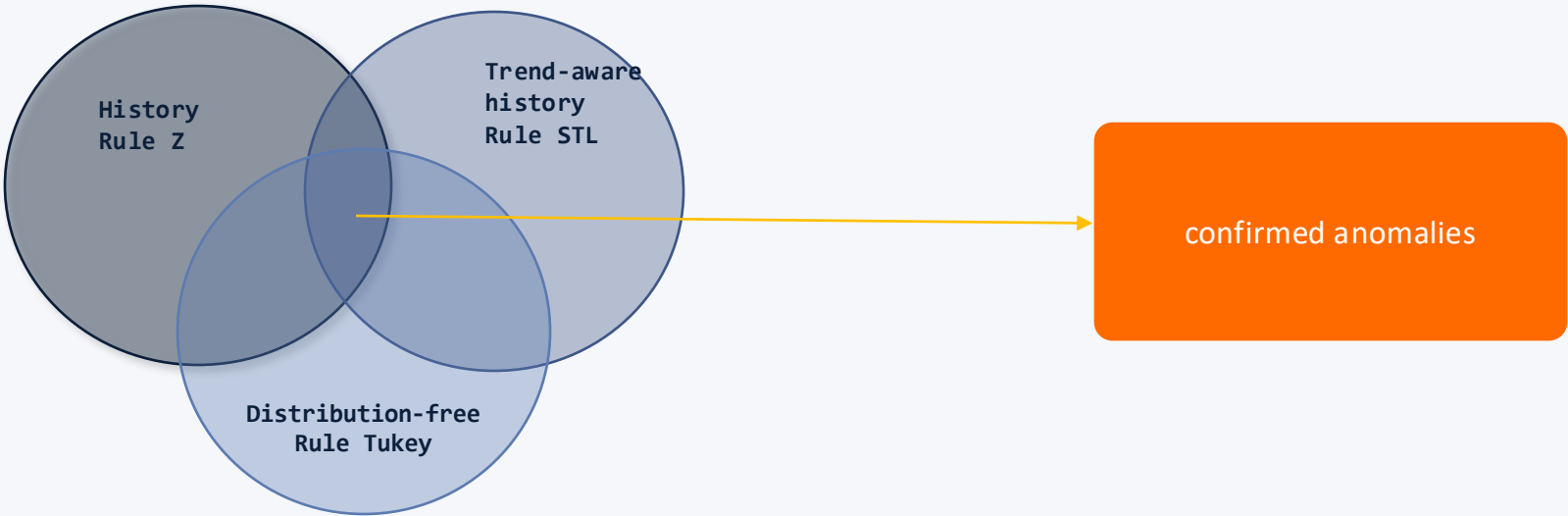


The metric moved $\geq 20\%$ in one hour.

e.g. EU eCPM \$3.25 → \$2.28 ($\Delta\% = -30\%$).

- ▶ volume gate: $\geq 10\text{K}$ requests global · $\geq 1\text{K}$ L1 · ≥ 500 L2
- ▶ no statistical calibration – pure magnitude threshold

either rule fires → candidate · three methods confirm it's *anomaly*



How fast? Fast enough for interactive RCA.

The User asks a question. We answer in the same breath.

“Is anything anomalous right now?”

2.9 s

▸ 71M rows · 6 slice grains × 5 metrics
· 35-day window

Enables: “What about Fridays?” — answered mid-incident, no re-run.

“Confirm it with 3 independent methods.”

3.5 s

▸ Rule Z + STL + Tukey · 141M rows ·
one SQL query

Enables: No batch job. Confirmation is on-demand, not a nightly artifact.

“Decompose one time-series with STL.”

53 ms

▸ seriesDecomposeSTL(840 points,
period=24)

Enables: All 78 slice-metric series decomposed in under 5 s.

“Drill into any segment on the fly.”

14 ms

▸ dictGet enrichment · 55K rows
returned from 398K scanned

Enables: Real-time slicing during Q&A. Every drill is instant.

Interactive during an incident. No 'wait for the next batch'.

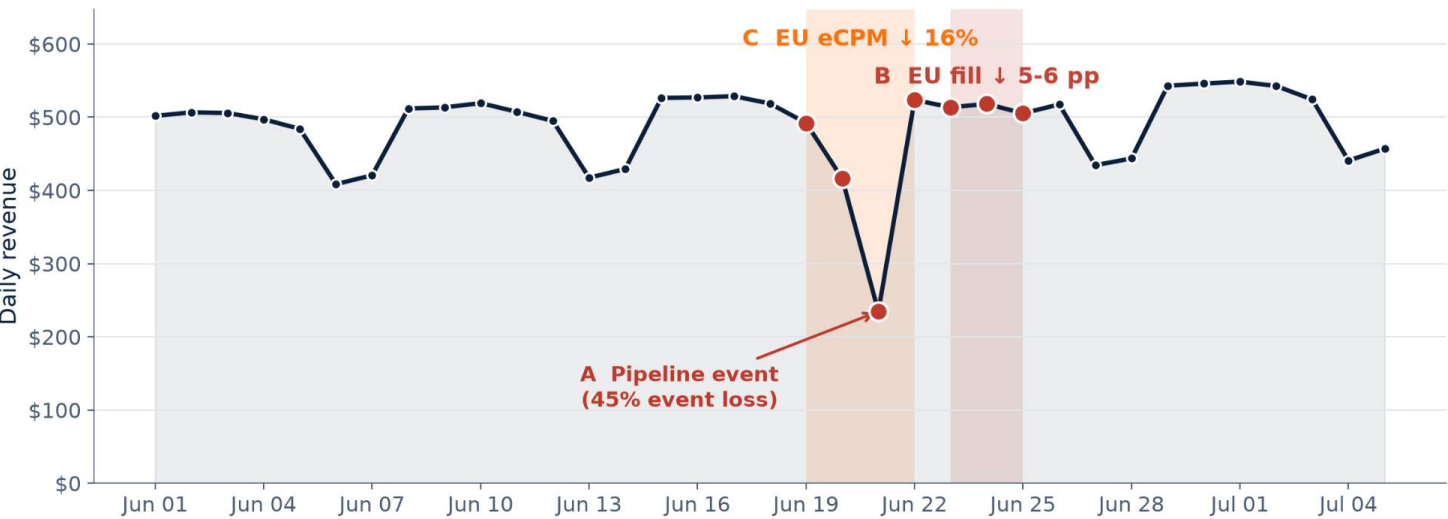
All timings measured against the live ClickHouse Cloud cluster, 2026-08-02.

PROOF — three incidents, one chart.

Global daily revenue, Jun 14 → Jun 28. Every number below is a live query.

Three incidents. All visible in one revenue chart.

Global daily revenue · 2026-06-01 → 2026-07-05 · every red dot survives triage



PROOF · ANOMALY A

The raw data was missing.

Sun Jun 14 · 225,383 events

Sun Jun 21 · 126,052 events (-44%)

Sun Jun 28 · 233,943 events

How the platform ruled it out.

$Revenue = Requests \times Fill \times eCPM \div 1000$

which factor moved on Jun 21?

revenue	-31 %	●	moved	
requests	-30 %	●	moved	← matches
revenue				
fill_rate	≈ 0 %	●	stable	
eCPM	≈ 0 %	●	stable	

→ Only volume moved. Identity balances.

→ Ingest event, ruled out — not a business incident.

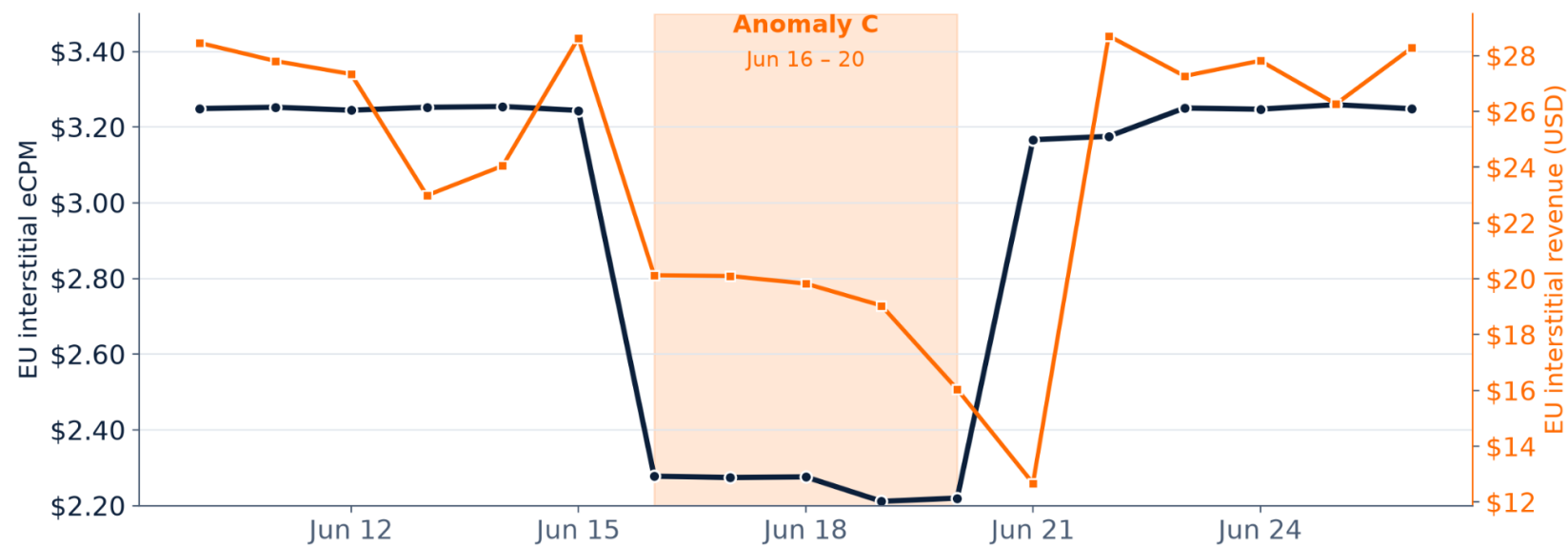
A, B, C all trip the rules. Only the revenue-identity walk exempts A — the deterministic rule-out the platform was built for.

PROOF — the sub-slice tells the truth.

Global eCPM barely moved. Drill to EU × interstitial and the drop is 10× larger.

Anomaly C — EU interstitial eCPM steps down, revenue tracks it.

5-day eCPM step-down of ~30% (\$3.25 → \$2.28). Uniform across every advertiser vertical.



Sub-slice drop

–30%

EU interstitial eCPM

Global view saw

–3%

why the drill matters

LLM ARCHITECTURE

Figure 1 — Closed-loop RCA (intended vs current)

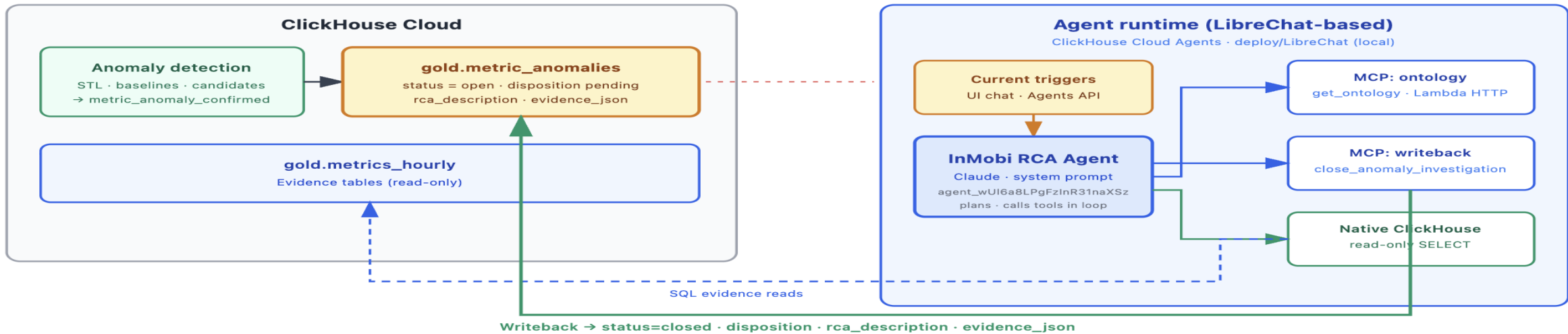


Figure 2 — Agent investigation loop (MCP calls per run)

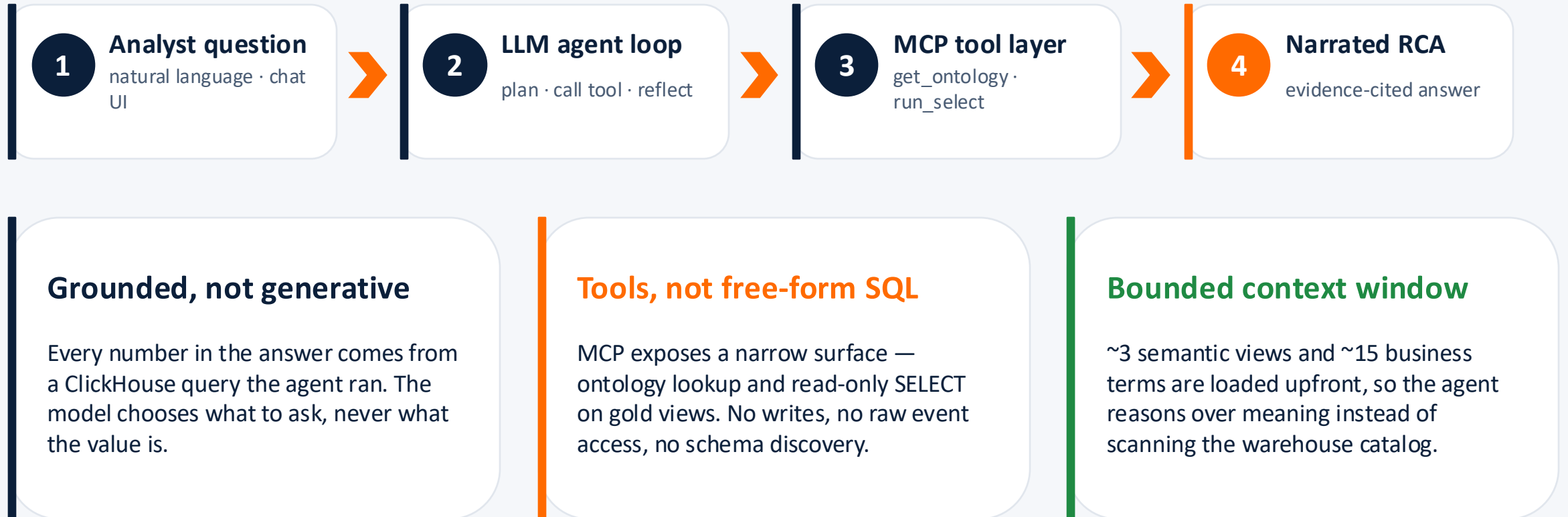


How MCP calls work inside the agent

- Agent Builder attaches MCP servers as Streamable HTTP endpoints (Bearer token auth).
- On each turn, the LLM planner decides which tool to call; the runtime POSTs JSON-RPC to the MCP URL.
- **get_ontology** returns bundled glossary_ontology.yaml (metrics, dimensions, constraints).
- ClickHouse evidence uses the **native** read-only ClickHouse capability (not a custom MCP in prod).
- **close_anomaly_investigation** runs ALTER UPDATE on gold.metric_anomalies (disposition required).
- Tool calls appear in the chat UI; there is no system.agents_* usage table for agent telemetry.
- Queue refresh does not yet auto-invoke the agent; use UI chat or Agents API.

LLM ARCHITECTURE — THE RCA AGENT

An LLM that reasons and narrates — never one that computes



GLOSSARY ONTOLOGY · MCP GET_ONTOLOGY

Precise context upfront — fewer SQL round trips

ClickHouse holds the numbers; the ontology holds the meaning — tables, metric decomposition, and SQL rules. Served via `glossary_ontology.yaml` (v1.3) through `get_ontology` MCP before any investigation SQL.

Without ontology

- ✗ Broad schema / column comments across many tables
- ✗ Invents metric formulas and rollups on the fly
- ✗ Raw events or wrong aggregations (avg vs sum-over-sum)
- ✗ SQL retries — more LLM turns per investigation

With ontology (our approach)

- ✓ First tool call: `get_ontology` — single bundled context
- ✓ Semantic views only — no raw bronze/silver/events
- ✓ Each term → `clickhouse_view` + column + rollup rule
- ✓ Relationships: revenue → requests, fill_rate, ecpm
- ✓ Constraints: hourly grain, sum-over-sum, filter order

SEMANTIC VIEWS → ONTOLOGY MAPPING

Anti-hallucination: SELECT only on curated gold views

Gold semantic views are pre-aggregated at business grain. The ontology maps each term to one view, column, and rollup rule — the agent does not invent formulas or discover schema at runtime.

Semantic view	Ontology maps	Agent runs (no ad-hoc math)
gold.metrics_hourly	Additive metrics, ratio formulas (sum-over-sum), slice dimensions	SELECT sum(revenue), sum(fills)/sum(requests) ... WHERE region, hour
gold.metric_hourly_by_slice	Pre-sliced rollups aligned with slice_type / slice_value	SELECT ... WHERE slice_type = ? AND slice_value = ?
gold.metric_anomalies	Queue fields, status, disposition, writeback columns	SELECT ... WHERE anomaly_id = ?

Blocked in ontology: bronze.* / silver.* events and gold.ad_events_semantic. Context stays small (~3 tables, ~15 business terms) — not full system.columns or warehouse-wide descriptions.

END-TO-END AUTOMATION · GOLD.METRIC_ANOMALIES

Automated RCA writeback in ClickHouse

The screenshot shows the ClickHouse web interface. At the top, there are tabs for 'metrics_h...', 'metric_an...', and 'metric_an...'. Below the tabs is a table named 'metric_anomalies'. The table has columns: #, status, incident_id, las..., investigat..., ale..., rca_descri..., and evidence_j... The first row is highlighted, showing an anomaly with status 'closed', incident_id 'NULL', and rca_description 'Global (all-region) interstitial'. To the right of the table is a 'Cell Inspector' panel showing the details of the selected cell. The details include a summary of the anomaly, a description of the issue, and a list of regions affected.

#	status	incident_id	las...	investigat...	ale...	rca_descri...	evidence_j...
1	closed	NULL		2026-08-02...		Global (all-region) interstitial	{"summary":...
2	closed	NULL		2026-08-02...		NAM native...	{"summary":...
3	closed	NULL		2026-08-02...		NAM blende...	{"summary":...

Cell Inspector

Global (all-region) interstitial

CTR rose ~124% at 2026-07-05 02:00 UTC - 0.0220 vs a 7-day same-hour baseline average of 0.0113 (a +94.8% delta by simple averaging, same direction/order of magnitude as the flagged +124%). The increase is format-specific: banner (0.0068), native (0.0101), and video (0.0123) CTR at the same hour on 07-05 were within their normal 8-day ranges, while interstitial CTR was the highest value seen in the 8-day window (previous range 0.006-0.014) and rewarded CTR was also elevated (0.030, near its 8-day high). Breaking interstitial down by region at 02:00 shows the increase was broad-based: APAC 0.0226, EU 0.0204, MEA 0.0204, NAM 0.0283 all above their ~01:00/03:00 neighbor-hour levels; only LATAM (62 impressions, 0 clicks) diverged, which is attributable to very low volume in that region/hour. Total volume (953 impressions, 21 clicks across 5 regions) is modest but the multi-region, format-specific

An anomaly in the queue triggers the agent:

- 1 it calls get_ontology
- 2 runs reproducible SELECTs on semantic views
- 3 ClickHouse computes every number

The LLM narrates only from that evidence — then writeback closes the row with rca_description and evidence_json automatically (status = closed).

Live from the cluster.

Numbers, straight from the pipeline.

■ **9,000,000**

events analysed · Jun 1 → Jul 5

■ **21,055**

critical candidates ·
gold.metric_anomaly_candidates

■ **3**

confirmed anomalies · triple /
double-agree

■ **5 × 6**

metrics × slice grains · every hour

Pulled from gold.metric_anomaly_confirmed on initially provided dataset.